Test Model on Multiple Datasets

This notebook evaluates a trained model on three test datasets:

- **Natural**: Uniformly random samples from [1, 10^13]
- **Cheat**: Numbers with prime factors only within the first 100 primes
- **Non-cheat**: Numbers with at least one prime factor outside the first 100 primes

It reports per-class performance for each dataset.

Configuration

Specify the model checkpoint and encoding to test.

IMPORTANT:

- `ENCODING` should be the **base encoding ONLY** (e.g., `interCRT100`, `CRT100`)
- Do **NOT** include dataset type suffixes like `_natural`, `_cheat`, or `_non_cheat`
- The notebook will automatically test on all three dataset types

```
Device detection: GPU not available
Using device: cpu

Testing model: ../models/model_interCRT100_cheat/mu/1/checkpoint.pth
Encoding: interCRT100
Task: mu
Results will be saved to: ../test_results/interCRT100_mu
```

Helper Functions

```
Helper functions loaded!
```

Check if Model Checkpoint Exists

```
✓ Checkpoint found: ../models/model_interCRT100_cheat/mu/1/checkpoint.pth
```

Check Test Data Files

```
Checking test data files:

✓ natural     : ../input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.tes
t (105.83 MB)
✗ cheat       : ../input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.test NO
T FOUND
✓ non_cheat   : ../input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.tx
t.test (105.82 MB)
```

Run Evaluation on Each Test Dataset

```
Evaluation function ready!
```

Evaluate on Natural Dataset

```
================================================================================
EVALUATING ON NATURAL DATASET
================================================================================
Running in CPU mode

Running command:
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/Gi
tHub/mobius_case_study/notebooks/../models/model_interCRT100_cheat/mu/1/checkpoint.pt
h --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/../input/inp
ut_dir_interCRT100_natural/mu_interCRT100_natural.txt.test --eval_size 10000 --data_t
ypes int[200]:range(-1,2) --operation data --num_workers...

================================================================================
Full output saved to: ../test_results/interCRT100_mu/eval_natural.log

✓ Evaluation completed successfully!

Overall Accuracy: 39.56%
```

Evaluate on Cheat Dataset

```
✗ Test file not found: ../input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.
test
```

Evaluate on Non-Cheat Dataset

```
================================================================================
EVALUATING ON NON-CHEAT DATASET
================================================================================
Running in CPU mode

Running command:
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/../models/model_interCRT100_cheat/mu/1/checkpoint.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/../input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --num_workers...

================================================================================
Full output saved to: ../test_results/interCRT100_mu/eval_non_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 39.49%
```

Compile Results

```
✓ Successfully evaluated on 2 dataset(s)

Results saved to: ../test_results/interCRT100_mu/all_results.json
```

Overall Performance Comparison

```
================================================================================
OVERALL PERFORMANCE SUMMARY
================================================================================
```

	Dataset	Accuracy (%)	Acc μ=0 (%)	Acc μ=1 (%)	Acc μ=-1 (%)	Perfect (%)	Correct (%)	XE Loss
0	Natural	39.56	99.923469	0.033772	1.218339	39.56	39.56	3.547213
1	Non-Cheat	39.49	100.000000	0.000000	0.885536	39.49	39.49	3.548577

```
Summary saved to: ../test_results/interCRT100_mu/overall_summary.csv
```

Precision, Recall, F1, and Confusion Matrix

Note: The Int2Int evaluator provides per-class accuracy but not the full confusion matrix. To compute precision, recall, and F1 properly, we need to run a custom evaluation that captures predictions vs ground truth.

The following section runs a custom evaluation to extract predictions and compute these metrics.

```
Classification metrics and parsing functions loaded!
```

Compute Classification Metrics from Evaluation Results

Use the predictions captured from the evaluation runs above:

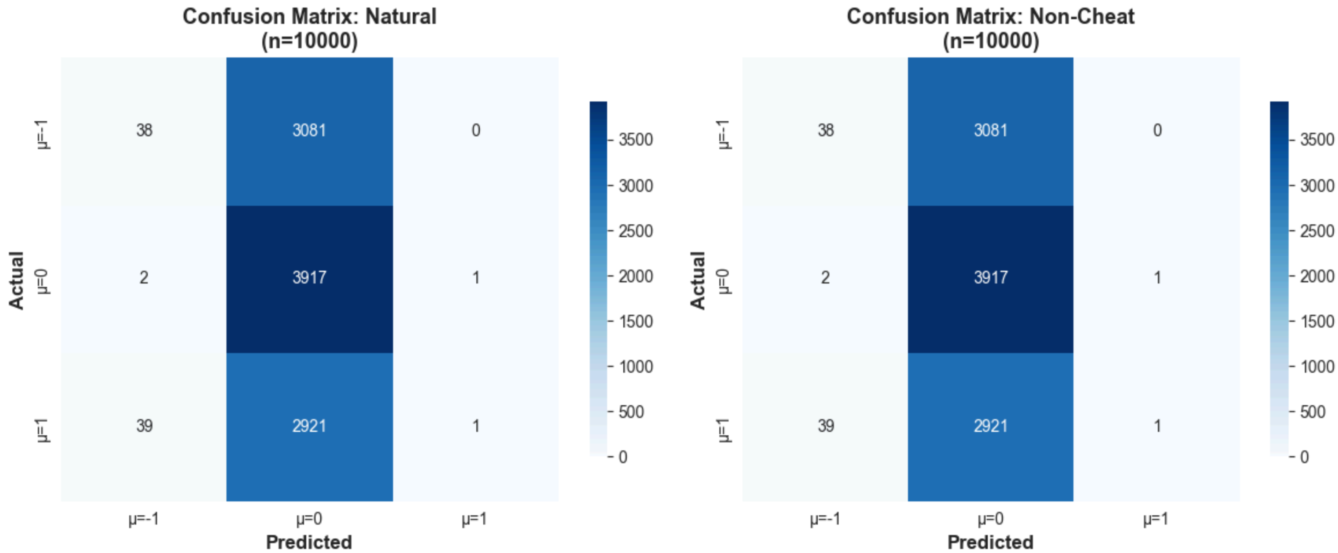
```
================================================================================
PARSING PREDICTIONS: NATURAL
================================================================================
Parsing predictions from: ../test_results/interCRT100_mu/eval_dump/debug/2x0fk2av6s/e
val.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!
Skipping cheat: no eval file available

================================================================================
PARSING PREDICTIONS: NON_CHEAT
================================================================================
Parsing predictions from: ../test_results/interCRT100_mu/eval_dump/debug/2x0fk2av6s/e
val.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

================================================================================
Classification metrics computed for 2 dataset(s)
================================================================================
```

Classification Report: Precision, Recall, F1

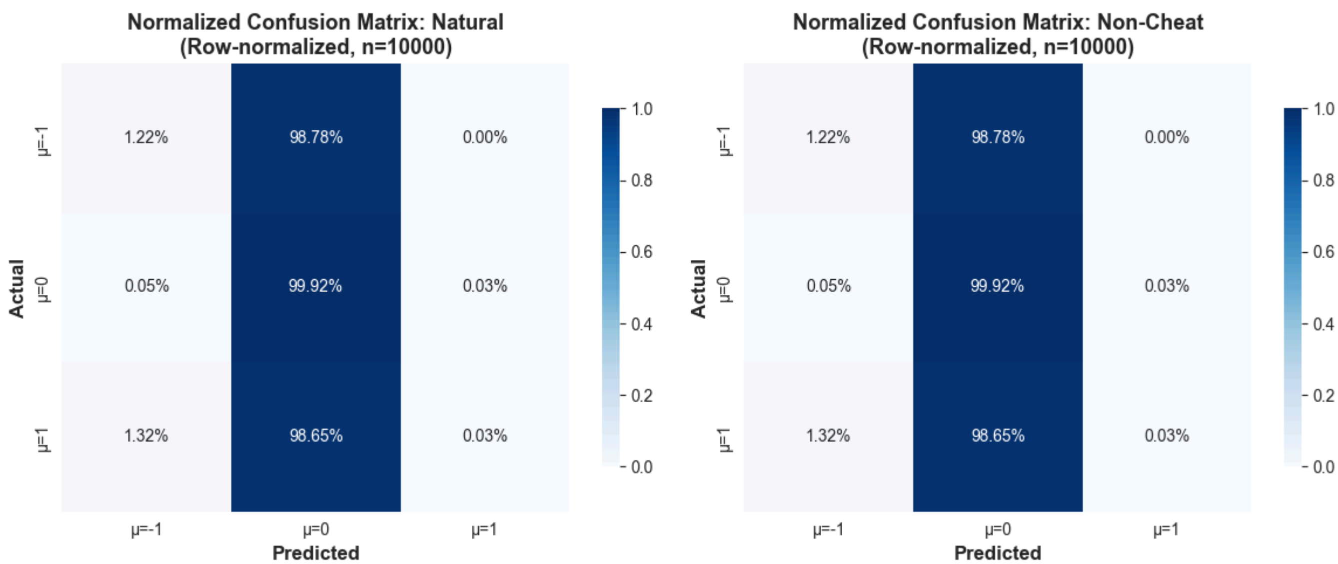
```
================================================================================
CLASSIFICATION REPORT: NATURAL
================================================================================

Samples evaluated: 10000
Invalid predictions: 0

              precision    recall  f1-score   support

        μ=-1       0.48      0.01      0.02      3119
         μ=0       0.39      1.00      0.57      3920
         μ=1       0.50      0.00      0.00      2961

    accuracy                           0.40     10000
   macro avg       0.46      0.34      0.20     10000
weighted avg       0.45      0.40      0.23     10000


Summary Metrics:
--------------------------------------------------
Metric                    Macro        Weighted
--------------------------------------------------
Precision                 0.4586       0.4529
Recall                    0.3373       0.3956
F1-Score                  0.1968       0.2295
--------------------------------------------------

================================================================================
CLASSIFICATION REPORT: NON_CHEAT
================================================================================

Samples evaluated: 10000
Invalid predictions: 0

              precision    recall  f1-score   support

        μ=-1       0.48      0.01      0.02      3119
         μ=0       0.39      1.00      0.57      3920
         μ=1       0.50      0.00      0.00      2961

    accuracy                           0.40     10000
   macro avg       0.46      0.34      0.20     10000
weighted avg       0.45      0.40      0.23     10000


Summary Metrics:
--------------------------------------------------
Metric                    Macro        Weighted
--------------------------------------------------
Precision                 0.4586       0.4529
Recall                    0.3373       0.3956
F1-Score                  0.1968       0.2295
--------------------------------------------------
```

Confusion Matrices

```
Figure saved to: ../test_results/interCRT100_mu/confusion_matrices.png
```




```
Figure saved to: ../test_results/interCRT100_mu/confusion_matrices_normalized.png
```



Per-Class Precision, Recall, F1

```
================================================================================
PER-CLASS METRICS: NATURAL
================================================================================
```

	Class	Precision	Recall	F1-Score
0	μ=-1	0.4810	0.0122	0.0238
1	μ=0	0.3949	0.9992	0.5661
2	μ=1	0.5000	0.0003	0.0007

```
Saved to: ../test_results/interCRT100_mu/per_class_metrics_natural.csv

================================================================================
PER-CLASS METRICS: NON_CHEAT
================================================================================
```

	Class	Precision	Recall	F1-Score
0	μ=-1	0.4810	0.0122	0.0238
1	μ=0	0.3949	0.9992	0.5661
2	μ=1	0.5000	0.0003	0.0007

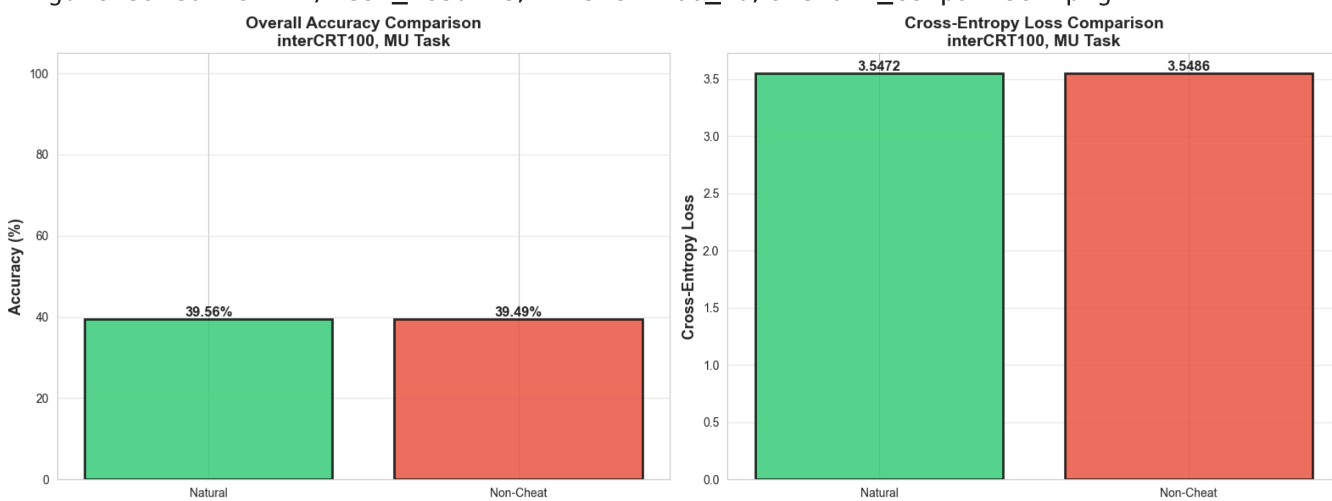
```
Saved to: ../test_results/interCRT100_mu/per_class_metrics_non_cheat.csv
```

Save All Detailed Metrics

```
All detailed metrics saved to: ../test_results/interCRT100_mu/detailed_classification
_metrics.json
Predictions saved to: ../test_results/interCRT100_mu/predictions_natural.npz
Predictions saved to: ../test_results/interCRT100_mu/predictions_non_cheat.npz
```

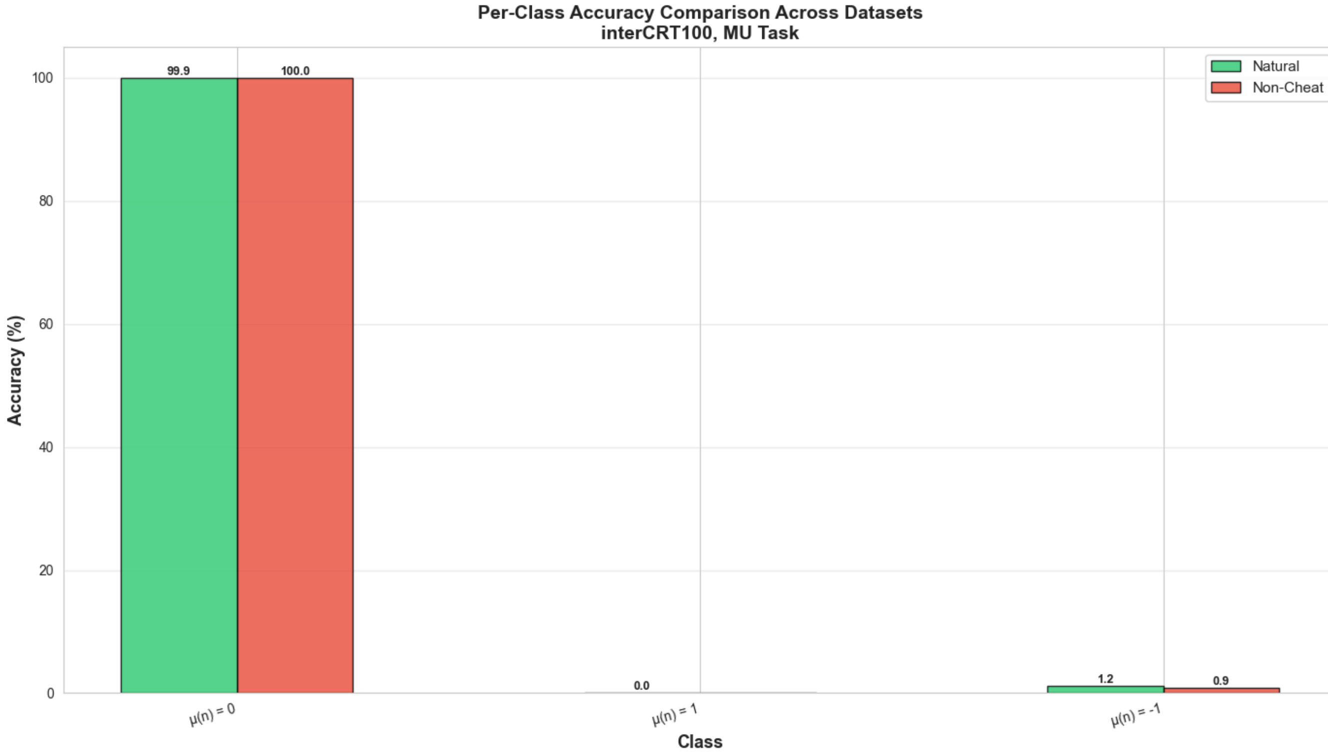
Visualization: Overall Performance Comparison

```
Figure saved to: ../test_results/interCRT100_mu/overall_comparison.png
```



Visualization: Per-Class Performance Comparison

```
Figure saved to: ../test_results/interCRT100_mu/per_class_comparison.png
```



Manual Inspection: Sample Predictions from Each Test Set

Randomly select 20 samples from each test dataset and display the number, true Mobius value, and model prediction.

```
natural: loaded 10000 numbers
cheat: test file not found
non_cheat: loaded 10000 numbers

================================================================================
===============
SAMPLE PREDICTIONS FROM EACH TEST SET
================================================================================
===============

================================================================================
===============
DATASET: NATURAL (showing 20 random samples)
================================================================================
===============
Correct: 11/20 (55.0%)
```

	n	True μ(n)	Predicted μ(n)	Correct
0	(sample 43)	0	0	Yes
1	(sample 269)	1	0	No
2	(sample 876)	1	0	No
3	(sample 1065)	0	0	Yes
4	(sample 1916)	1	0	No
5	(sample 2922)	0	0	Yes
6	(sample 3299)	-1	0	No
7	(sample 3439)	0	0	Yes
8	(sample 4090)	0	0	Yes
9	(sample 5138)	0	0	Yes
10	(sample 5786)	1	0	No
11	(sample 6255)	0	0	Yes
12	(sample 6821)	-1	0	No
13	(sample 7768)	-1	0	No
14	(sample 8009)	1	0	No
15	(sample 8146)	0	0	Yes
16	(sample 8439)	0	0	Yes
17	(sample 9443)	1	0	No
18	(sample 9825)	0	0	Yes
19	(sample 9885)	0	0	Yes

```
Skipping cheat: no predictions available

================================================================================
===============
DATASET: NON_CHEAT (showing 20 random samples)
================================================================================
===============
Correct: 5/20 (25.0%)
```

	n	True μ(n)	Predicted μ(n)	Correct
0	(sample 336)	1	0	No
1	(sample 3457)	-1	0	No
2	(sample 3662)	1	0	No
3	(sample 5862)	1	0	No
4	(sample 6001)	1	0	No
5	(sample 6109)	0	0	Yes
6	(sample 6421)	1	0	No
7	(sample 6676)	1	0	No
8	(sample 6971)	-1	0	No
9	(sample 6995)	-1	0	No
10	(sample 7115)	0	0	Yes
11	(sample 7247)	1	0	No
12	(sample 7769)	-1	0	No
13	(sample 8492)	1	0	No
14	(sample 8844)	0	0	Yes
15	(sample 9758)	0	0	Yes
16	(sample 9784)	0	0	Yes
17	(sample 9834)	-1	0	No
18	(sample 9925)	1	0	No
19	(sample 9982)	-1	0	No

==
==============